

A QPINN Framework with Quantum Trainable Embeddings for the Lid-Driven Cavity Problem

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Abstract—The steady incompressible Navier–Stokes equations pose significant computational challenges due to their nonlinear convective terms and pressure–velocity coupling. Physics-informed neural networks (PINNs) provide a mesh-free framework for approximating such systems, but classical PINNs can experience optimization difficulties in nonlinear flow regimes. In this work, we propose a quantum physics-informed neural network (QPINN) framework with a quantum neural network (QNN)-based trainable embedding for the lid-driven cavity problem. The proposed approach uses a QNN to learn data-adaptive quantum feature maps that encode spatial coordinates before they are processed by a variational quantum circuit within a physics-informed loss formulation. Numerical experiments show that the proposed QNN-TE-QPINN exhibits stable training behavior and competitive solution accuracy compared with classical PINNs and hybrid quantum models using classical embeddings, while requiring significantly fewer trainable parameters. Rather than claiming computational speedup, these results highlight the potential of trainable quantum embeddings for parameter-efficient physics-informed learning. The findings suggest that embedding design plays an important role in quantum-assisted PDE solvers and support further investigation of QNN-based trainable embeddings for nonlinear fluid dynamics benchmarks.

Index Terms—Quantum machine learning, quantum physics-informed neural networks, and partial differential equations.

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I. INTRODUCTION

The numerical solution of nonlinear partial differential equations (PDEs) remains one of the central challenges in scientific computing. Among these equations, the incompressible Navier–Stokes equations play a fundamental role in modeling fluid motion across science and engineering, yet their nonlinear convective structure and strong pressure–velocity coupling make them particularly difficult to solve accurately. The lid-driven cavity flow is one of the most widely studied benchmark problems in computational fluid dynamics (CFD), and it has long served as a standard test case for assessing the robustness, accuracy, and efficiency of numerical solvers for incompressible flow. As demonstrated in the classical benchmark study of Ghia, Ghia, and Shin [1] and in subsequent investigations, the square cavity geometry is simple, yet the resulting flow exhibits sufficiently rich dynamics to pose a demanding challenge for numerical methods.

A primary source of difficulty in solving the lid-driven cavity problem arises from the nonlinear convection terms in the Navier–Stokes equations, which introduce self-interaction within the velocity field and lead to complex flow features such as vortices and boundary-layer formation [2]. Moreover, no closed-form analytical solution exists for this problem, and accurate simulation typically requires fine spatial discretization, particularly at higher Reynolds numbers, where multiple vortical structures develop. These computational demands motivate the exploration of alternative approximation frameworks that can capture nonlinear flow behavior without relying exclusively on conventional mesh-based techniques.

Physics-informed neural networks (PINNs), introduced by Raissi, Perdikaris, and Karniadakis [3], provide one such

framework by embedding governing equations, boundary conditions, and initial conditions directly into the loss function of a neural network. In this manner, PINNs learn continuous surrogate models of PDE solutions while avoiding explicit discretization of the spatial domain. Nevertheless, despite their conceptual elegance, classical PINNs often encounter optimization challenges in strongly nonlinear and multiscale settings, including slow convergence, sensitivity to loss balancing, and difficulty in resolving complex solution manifolds.

These limitations have motivated interest in hybrid quantum–classical approaches to physics-informed learning. In the noisy intermediate-scale quantum (NISQ) era, variational quantum algorithms based on parameterized quantum circuits have emerged as a practical paradigm for near-term quantum computing [4]. Such approaches offer access to high-dimensional Hilbert-space representations while remaining compatible with classical optimization loops. Building on this idea, Kyriienko, Paine, and Elfving demonstrated that differentiable quantum circuits can be trained to satisfy nonlinear differential equations and boundary conditions, thereby opening a path toward quantum-assisted PDE solvers [5].

A key architectural component in quantum-assisted PINNs is the embedding mechanism that maps classical input coordinates into quantum states prior to variational processing. In the original trainable-embedding QPINN framework, Berger, Hosters, and Möller [6] introduced adaptive, trainable feature maps to overcome the expressivity limitations of fixed encodings, showing that learned embeddings can improve solution accuracy while reducing qubit requirements. Motivated by this insight, our recent work has systematically investigated trainable embeddings in quantum-assisted physics-informed learning, examining both classical and fully quantum embedding strategies and their impact on convergence, gradient structure, parameter scaling, and solution quality in nonlinear reaction–diffusion systems [7]–[9].

In the present study, we extend this line of research to nonlinear fluid dynamics by proposing a QPINN framework with a quantum neural network (QNN)-based trainable embedding for the lid-driven cavity problem. In this architecture, a dedicated QNN learns a mapping from collocation point coordinates to quantum data-encoding angles, which are then processed by a variational quantum circuit to approximate the solution fields. Unlike fixed or handcrafted encodings, the proposed approach learns the coordinate-to-angle mapping jointly with the PDE solver. This gives the quantum circuit an adaptive input representation that can better match the spatial structure of the lid-driven cavity flow.

Contributions and Scope: The main contributions of this work are as follows. First, we formulate a trainable-embedding quantum physics-informed neural network (QPINN) framework for the steady incompressible Navier–Stokes equations governing the lid-driven cavity problem. Second, we introduce a QNN-based trainable embedding mechanism that learns a coordinate-to-angle encoding transformation jointly with the variational quantum solver. Third, we provide an empirical comparison with classical PINNs and hybrid QPINN baselines

employing alternative embedding strategies. Finally, through numerical experiments, we show that embedding design is an important factor in quantum-assisted PDE solvers, extending the use of trainable-embedding QPINNs to a nonlinear fluid-flow benchmark.

We do not claim a complexity-theoretic quantum advantage for the proposed method. Instead, the observed benefit is a reduction in the number of trainable parameters required to achieve comparable accuracy. In this sense, parameter efficiency means that the model has fewer degrees of freedom to optimize, which can reduce memory requirements, lower optimizer overhead, and make the training problem more compact. In the reported experiments, the proposed QNN-TE-QPINN achieves competitive accuracy for the lid-driven cavity problem using approximately 360 trainable parameters, compared with approximately 6,600 parameters in the classical PINN baseline. This comparison should therefore be understood as an empirical reduction in model size and trainable degrees of freedom, rather than as a demonstrated computational speedup.

II. NAVIER–STOKES EQUATIONS IN STREAM-FUNCTION FORMULATION

In this work, we consider the steady two-dimensional incompressible Navier–Stokes equations for the lid-driven cavity problem on the square domain $\Omega = [0, 1] \times [0, 1]$. Let $u(x, y)$ and $v(x, y)$ denote the horizontal and vertical velocity components, respectively, and let $p(x, y)$ denote the pressure field. In the nondimensional steady-state setting considered here, the momentum equations are given by

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \quad (1)$$

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{Re} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right), \quad (2)$$

where Re denotes the Reynolds number.

A major difficulty in solving the Navier–Stokes equations arises from the nonlinear convective terms $uu_x + vv_y$ and $uv_x + vv_y$, where subscripts denote partial derivatives, e.g., $u_x = \partial u / \partial x$ and $v_y = \partial v / \partial y$. These nonlinear self-advection terms couple the unknown velocity field to its own spatial derivatives. This nonlinear self-advection is responsible for the rich behavior of incompressible flows, including vortex formation and increased sensitivity to boundary conditions as the Reynolds number increases. For two-dimensional incompressible flows on a simply connected domain, the divergence-free condition implies the existence of a scalar stream function $\psi(x, y)$. Therefore, we adopt the stream-function formulation

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (3)$$

With this definition, the divergence-free condition is satisfied identically:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} = 0. \quad (4)$$

Therefore, instead of learning the velocity components directly, the model approximates the pair (p, ψ) , and the velocity field is recovered through (3). This is the formulation used in the implementation, where incompressibility is enforced implicitly through the stream function rather than through a separate continuity residual in the loss function.

Using (3), the momentum equations are enforced in residual form as

$$\mathcal{R}_x := u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + \frac{\partial p}{\partial x} - \frac{1}{Re} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = 0, \quad (5)$$

$$\mathcal{R}_y := u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + \frac{\partial p}{\partial y} - \frac{1}{Re} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) = 0, \quad (6)$$

where u and v are obtained from derivatives of ψ .

For the lid-driven cavity problem, the boundary conditions are defined as follows. On the top boundary, the lid moves with unit horizontal velocity and zero vertical velocity:

$$u(x, 1) = 1, \quad v(x, 1) = 0. \quad (7)$$

On the remaining three walls, define $\partial\Omega_{\text{wall}} := \{(x, y) \in \Omega \mid x = 0 \text{ or } x = 1 \text{ or } y = 0\}$. The no-slip boundary conditions are imposed as

$$u = 0, \quad v = 0, \quad \text{on } \partial\Omega_{\text{wall}}. \quad (8)$$

Since pressure is determined only up to an additive constant in incompressible flow, a reference pressure is prescribed to ensure uniqueness. In this work, we impose

$$p(0, 0) = 0. \quad (9)$$

Hence, the learning problem consists of approximating $p(x, y)$ and $\psi(x, y)$ such that the induced velocity field from (3) satisfies the momentum residuals (5)–(6), together with the lid-driven and no-slip boundary conditions (7)–(8) and the reference pressure condition (9). This formulation matches the computational structure used in our solver.

III. QUANTUM PHYSICS-INFORMED LEARNING FRAMEWORK

In this section, we introduce the hybrid quantum–classical physics-informed learning framework used to approximate the steady incompressible Navier–Stokes equations in the lid-driven cavity problem. The proposed approach extends the trainable-embedding quantum PINN paradigm to nonlinear fluid dynamics by combining (i) quantum neural network (QNN)-based embeddings, (ii) a hardware-efficient variational quantum circuit, and (iii) a physics-informed loss that enforces momentum balance, boundary conditions, and pressure normalization. Figure 1 provides an overview of the computational pipeline.

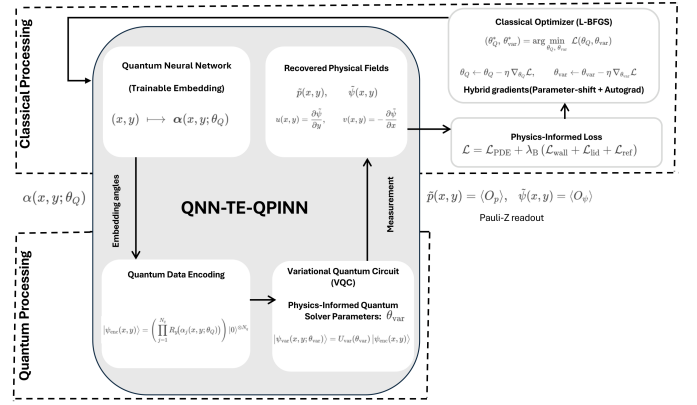


Fig. 1. Architecture of the proposed QNN-TE-QPINN framework. Classical processing maps spatial collocation points (x, y) to trainable embedding parameters $\alpha(x, y; \theta_Q)$ via a quantum neural network. These parameters are used for quantum data encoding, after which a variational quantum circuit acts as the physics-informed quantum solver to generate a variational quantum state. Observable expectation values obtained through Pauli–Z measurements yield predictions for the pressure $p(x, y)$ and stream function $\psi(x, y)$, from which velocity fields are recovered. Physics-informed PDE residuals, boundary constraints, and reference conditions are enforced through a classical loss function. All parameters are updated using a classical L-BFGS optimizer with hybrid gradients computed via automatic differentiation and parameter-shift rules.

A. Trainable Quantum Embedding Network (QNN-Based)

To encode the spatial coordinates (x, y) of the cavity domain into a form suitable for quantum processing, we employ a trainable quantum embedding network. This module translates classical inputs into a set of quantum features that determine how the data enter the variational quantum circuit, and it provides the only channel through which the physical coordinates influence the quantum model. As a result, the embedding circuit defines a data-dependent quantum feature map whose structure is learned jointly with the PDE solution.

Before being embedded, the coordinates are rescaled through an affine mapping,

$$\tilde{x} = \mathcal{N}(x), \quad \tilde{y} = \mathcal{N}(y), \quad (10)$$

which maps the original coordinate range to the normalized interval $[-1, 1]$. This normalization balances the magnitudes of the two spatial dimensions and improves the numerical stability of the hybrid optimization process. The normalized coordinates $(\tilde{x}, \tilde{y})$ are then processed by a parameterized quantum circuit equipped with variational parameters θ_Q . Acting on the reference state $|0\rangle^{\otimes N_q}$, this circuit produces the embedded quantum state

$$|\psi_{\text{embed}}(\tilde{x}, \tilde{y}; \theta_Q)\rangle = U_{\text{embed}}(\tilde{x}, \tilde{y}; \theta_Q)|0\rangle^{\otimes N_q}. \quad (11)$$

The unitary U_{embed} consists of alternating layers of input-dependent gates—through which the coordinates affect the state—and trainable rotation layers parameterized by θ_Q , along with a predefined pattern of entangling gates, such as nearest-neighbor CNOT gates. This design enables the embedding map to adapt its representation during training and to distribute

coordinate information across the qubits in a highly nonlinear manner.

To convert the embedded quantum state into a usable feature vector, we evaluate a prescribed set of Hermitian observables $\{O_i\}_{i=1}^{N_q}$, obtaining

$$\Gamma(\tilde{x}, \tilde{y}) = [\alpha_1(\tilde{x}, \tilde{y}), \alpha_2(\tilde{x}, \tilde{y}), \dots, \alpha_{N_q}(\tilde{x}, \tilde{y})]^\top, \quad (12)$$

where $\alpha_i(\tilde{x}, \tilde{y}) = g_i(\langle \psi_{\text{embed}}(\tilde{x}, \tilde{y}; \theta_Q) | O_i | \psi_{\text{embed}}(\tilde{x}, \tilde{y}; \theta_Q) \rangle)$, and $g_i(\cdot)$ is a scaling function that maps the measurement outcome to a valid rotation angle. In this work, we take O_i to be the Pauli- Z operator on the i -th qubit, and the post-processing function reduces to a simple linear scaling $g_i(s) = \pi s$, which maps the expectation values in $[-1, 1]$ to rotation angles used in the quantum encoding unitary. This choice provides a simple yet expressive quantum feature map, since Pauli- Z measurements yield bounded outputs in $[-1, 1]$, and the linear scaling ensures that the embedding angles span the full rotation range required for downstream encoding. Therefore, the QNN embedding learns a nonlinear coordinate transformation tailored to the flow physics, enabling the quantum model to represent vorticity, shear, and pressure-velocity coupling using a compact and hardware-efficient architecture.

Quantum encoding operator: The embedding circuit produces classicalized quantum features, which are then re-encoded into the solver circuit through a standard rotational data-encoding unitary. The angles α_i determine the data-encoding unitary applied prior to the variational circuit,

$$U_{\text{enc}}(\tilde{x}, \tilde{y}; \theta_Q) = \bigotimes_{i=1}^{N_q} R_y(\alpha_i(\tilde{x}, \tilde{y})), \quad (13)$$

where $R_y(\cdot) = \exp[-i(\cdot)\sigma_y/2]$ denotes a single-qubit rotation about the y -axis. This operator produces the encoded input-dependent quantum state

$$|\psi_{\text{enc}}(\tilde{x}, \tilde{y})\rangle = U_{\text{enc}}(\tilde{x}, \tilde{y}; \theta_Q)|0\rangle^{\otimes N_q}. \quad (14)$$

Because the embedding parameters θ_Q are optimized jointly with the variational parameters of the downstream circuit, the feature map $\Gamma(\tilde{x}, \tilde{y})$ adapts dynamically to the structure of the Navier-Stokes residuals. This co-training strategy enables the embedding to learn representations that emphasize key flow characteristics—such as vorticity concentration, shear layers, and coupled pressure-velocity behavior—while the use of shallow, hardware-efficient quantum circuits ensures compatibility with near-term devices. The result is a flexible and fully quantum trainable embedding mechanism that provides a consistent representation of the domain and can enhance the expressiveness of the overall QPINN model.

B. Variational Quantum Circuit Design

After the embedding module produces an input-dependent quantum state, this state is passed through a parameterized variational quantum circuit (VQC), which forms the trainable quantum component of the hybrid learning architecture. The purpose of the VQC is to transform the encoded state into a

latent quantum representation from which the physical output fields can be obtained through measurement.

The variational circuit is governed by a set of trainable parameters θ_{var} and is applied uniformly to all spatial inputs. That is, the same variational unitary acts repeatedly on quantum states generated from different coordinate-dependent embeddings. This parameter-sharing strategy lowers the number of trainable quantum variables and enforces a single latent representation for the predicted Navier-Stokes fields. Distinct physical quantities may then be extracted from the resulting quantum state by measuring different observables.

The variational unitary is constructed as a sequence of L layers, $U_{\text{var}}(\theta_{\text{var}}) = U_L(\theta_L) \cdots U_2(\theta_2) U_1(\theta_1)$, and each layer $U_\ell(\theta_\ell)$ consists of a product of parameterized and fixed operations. More precisely, each trainable block is given by

$$U_\ell(\theta_\ell) = \prod_m \exp(-i\theta_{\ell,m} H_m) W_m, \quad (15)$$

where the generators H_m are predetermined Hermitian operators and the operators W_m denote fixed, non-parameterized unitaries.

Each layer thus contains learnable single-qubit rotations, followed by a fixed entangling pattern. In typical hardware-efficient designs, the generators H_m are chosen from the Pauli basis and correspond to rotations about the coordinate axes of the Bloch sphere, while the operators W_m implement a nearest-neighbor entangling structure such as a sequence of CNOT gates. This layered composition allows the expressive power of the circuit to grow with depth while preserving a shallow architecture suitable for near-term quantum devices.

For a given spatial location (x, y) , the variational quantum state is obtained by applying the shared VQC to the encoded quantum state generated by the data-encoding operator:

$$|\psi_{\text{var}}(x, y; \Theta)\rangle = U_{\text{var}}(\theta_{\text{var}}) U_{\text{enc}}(\tilde{x}, \tilde{y}; \theta_Q) |0\rangle^{\otimes N_q}, \quad (16)$$

where $\Theta = (\theta_Q, \theta_{\text{var}})$ collects all trainable parameters of the model.

The predicted pressure and stream-function fields are then obtained as expectation values of the corresponding observables evaluated on the state $|\psi_{\text{var}}\rangle$. The variational parameters are optimized simultaneously with the embedding parameters using the physics-informed loss introduced in the following subsection. Repeated application of the same VQC to quantum states encoding different points in the domain enables the model to capture the nonlinear interactions, spatial gradients, and flow structures characteristic of incompressible Navier-Stokes dynamics, while maintaining a hardware-efficient quantum circuit compatible with near-term implementations.

C. Quantum Readout and Observable Design

The hybrid quantum model produces predictions for the pressure and stream-function fields by evaluating expectation values of suitably chosen Hermitian observables acting on the variational quantum state. While the embedding and variational circuits generate a latent quantum representation of the

spatial coordinates, distinct physical fields are extracted at the readout stage through field-specific observable operators.

For each output field $i \in \{p, \psi\}$, we assign a corresponding readout operator O_i acting on the N_q -qubit Hilbert space. In this work, we employ Pauli- Z -based observables of the form $O_i = \sum_{j=1}^{N_q} Z_j$, where Z_j denotes the Pauli- Z operator acting on the j -th qubit. This choice yields a single scalar expectation value per circuit evaluation, providing an efficient and hardware-friendly mechanism for mapping quantum states to classical outputs. Pauli- Z observables are particularly advantageous in physics-informed learning, as their bounded spectrum leads to stable expectation values. This boundedness improves the numerical conditioning of the PDE residuals and their derivatives, which in turn mitigates training instabilities and oscillatory behavior during optimization. The predicted pressure and stream-function fields are therefore computed as

$$\tilde{p}(x, y) = \langle \psi_{\text{var}}(x, y; \Theta) | O_p | \psi_{\text{var}}(x, y; \Theta) \rangle, \quad (17)$$

$$\tilde{\psi}(x, y) = \langle \psi_{\text{var}}(x, y; \Theta) | O_\psi | \psi_{\text{var}}(x, y; \Theta) \rangle, \quad (18)$$

where $|\psi_{\text{var}}(x, y; \Theta)\rangle$ denotes the variational quantum state produced by the embedding and variational circuits, and O_p and O_ψ are readout operators associated with the pressure and stream-function fields, respectively. The specific solver architecture adopted in the experiments—including whether readouts are obtained from a shared variational circuit or from independent solver instances—is detailed in the simulation section. While Pauli- Z readouts provide a robust and low-overhead extraction strategy, the proposed framework readily supports more general Hermitian observables, such as weighted Pauli combinations or problem-dependent operators, provided they remain efficiently measurable on near-term quantum hardware. The choice of readout operator thus reflects a balance between expressiveness, numerical stability, and practical feasibility in quantum-assisted solutions of the incompressible Navier-Stokes equations.

D. Physics-Informed Loss Function

The trainable parameters of the QNN-TE-QPINN model are optimized by minimizing a physics-informed loss function that enforces the steady incompressible Navier-Stokes equations together with the associated boundary and reference constraints. The loss is composed of four terms: the interior momentum residuals, the no-slip wall conditions, the lid-driven boundary condition, and a reference pressure constraint.

PDE residual loss: At interior collocation points, the model predicts the pressure and stream-function fields (p, ψ) , from which the velocity components are recovered via $u = \partial\psi/\partial y$ and $v = -\partial\psi/\partial x$. Substituting these expressions into the momentum equations yields the residuals

$$\mathcal{R}_x(x, y) = u u_x + v u_y + p_x - \frac{1}{Re} (u_{xx} + u_{yy}), \quad (19)$$

$$\mathcal{R}_y(x, y) = u v_x + v v_y + p_y - \frac{1}{Re} (v_{xx} + v_{yy}), \quad (20)$$

where all spatial derivatives are computed using automatic differentiation through the hybrid quantum-classical model.

The interior physics loss is defined as the mean squared magnitude of the momentum residuals,

$$\mathcal{L}_{\text{PDE}} = \frac{1}{|\Omega_{\text{int}}|} \sum_{(x, y) \in \Omega_{\text{int}}} (\mathcal{R}_x(x, y)^2 + \mathcal{R}_y(x, y)^2). \quad (21)$$

Boundary losses: The no-slip boundary condition requires $u = 0$ and $v = 0$ on the stationary walls, while the moving lid enforces $u = 1$ and $v = 0$ at $y = 1$. These constraints are imposed weakly by penalizing the squared deviations of the predicted velocities at the corresponding boundary collocation points,

$$\mathcal{L}_{\text{wall}} = \frac{1}{|\partial\Omega_{\text{wall}}|} \sum_{(x, y) \in \partial\Omega_{\text{wall}}} (u(x, y)^2 + v(x, y)^2), \quad (22)$$

$$\mathcal{L}_{\text{lid}} = \frac{1}{|\partial\Omega_{\text{lid}}|} \sum_{(x, y) \in \partial\Omega_{\text{lid}}} ((u(x, y) - 1)^2 + v(x, y)^2). \quad (23)$$

Reference pressure constraint: Since the pressure field in incompressible flow is defined only up to an additive constant, a reference value is prescribed to ensure uniqueness. In this work, we enforce the normalization condition $p(0, 0) = 0$ through the penalty term

$$\mathcal{L}_{\text{ref}} = p(0, 0)^2. \quad (24)$$

Total loss: The overall physics-informed loss combines the interior residual with the boundary and reference terms. To balance the relative contributions of the different constraints during optimization, the boundary and reference losses are scaled by a weighting parameter λ_B . The total loss is given by $\mathcal{L} = \mathcal{L}_{\text{PDE}} + \lambda_B (\mathcal{L}_{\text{wall}} + \mathcal{L}_{\text{lid}} + \mathcal{L}_{\text{ref}})$. By jointly minimizing $\mathcal{L}$ with respect to all trainable parameters Θ , the hybrid quantum model learns pressure and stream-function fields that satisfy the Navier-Stokes equations while remaining consistent with the lid-driven cavity boundary conditions.

E. Gradient Structure and Derivative Computation

Training the QNN-TE-QPINN model requires computing derivatives of the predicted pressure and stream-function fields with respect to both the spatial coordinates and the trainable parameters of the hybrid architecture. These derivatives enter directly into the evaluation of the momentum residuals, which involve first- and second-order spatial derivatives of the velocity components. As a result, the learning procedure depends on an interplay between quantum differentiation of the embedding and variational circuits and classical automatic differentiation of the PDE operators.

Since the predicted fields $\tilde{p}(x, y)$ and $\tilde{\psi}(x, y)$ are obtained as expectation values of the variational quantum state, their dependence on (x, y) arises exclusively through the data-encoding angles generated by the quantum embedding network. By the chain rule, spatial derivatives decompose into

$$\frac{\partial \tilde{f}}{\partial x} = \sum_{m=1}^{N_q} \frac{\partial \tilde{f}}{\partial \alpha_m} \frac{\partial \alpha_m}{\partial x}, \quad f \in \{p, \psi\}, \quad (25)$$

where $\partial\tilde{f}/\partial\alpha_m$ denotes the sensitivity of the quantum expectation value with respect to the rotation angle on the m -th qubit, and $\partial\alpha_m/\partial x$ is obtained by classical backpropagation through the QNN embedding network. An analogous expression holds for $\partial\tilde{f}/\partial y$. Second-order derivatives required by the Navier–Stokes operators ($u_{xx}, u_{yy}, v_{xx}, v_{yy}$) follow from repeated application of the same chain rule and are computed through nested automatic differentiation.

The gradients of the total loss with respect to any variational or embedding parameter $\theta \in \Theta$ take the form

$$\frac{\partial\mathcal{L}}{\partial\theta} = \sum_{(x,y) \in \Omega_{\text{int}}} \left(\frac{\partial\mathcal{L}}{\partial\tilde{p}} \frac{\partial\tilde{p}}{\partial\theta} + \frac{\partial\mathcal{L}}{\partial\tilde{\psi}} \frac{\partial\tilde{\psi}}{\partial\theta} \right) + \frac{\partial\mathcal{L}_{\text{B}}}{\partial\theta}, \quad (26)$$

where $\mathcal{L}_{\text{B}}$ denotes the collection of boundary and reference terms. The derivatives $\partial\tilde{p}/\partial\theta$ and $\partial\tilde{\psi}/\partial\theta$ reduce to gradients of quantum expectation values with respect to circuit parameters and are evaluated through the differentiable quantum-classical interface. In the present implementation, these gradients are computed using the automatic differentiation capabilities of the underlying hybrid framework, which internally applies parameter-shift rules for quantum gates and classical backpropagation for the embedding network.

Because the momentum residuals admit a closed-form dependence on the predicted fields and their spatial derivatives, the derivatives $\partial\mathcal{L}/\partial\tilde{p}$ and $\partial\mathcal{L}/\partial\tilde{\psi}$ are obtained directly via automatic differentiation. The resulting optimization pipeline therefore combines classical autograd for the PDE components with quantum-aware gradient evaluation for the circuit parameters, enabling the hybrid model to capture the nonlinear coupling, steep gradients, and boundary layer behavior characteristic of the lid-driven cavity flow.

F. Computational Cost and Parameter Scaling

The training cost of the QNN-TE-QPINN architecture arises from a combination of classical and quantum components, each contributing distinctly to the overall computational load. For each training iteration, the dominant expense comes from evaluating the hybrid model at all collocation points and computing the required gradients through both the embedding network and the variational circuit.

On the quantum side, the cost scales with the number of qubits N_q and the depth L of the variational circuit. Each trainable rotation gate introduces a parameter whose gradient must be computed through quantum-aware differentiation, and thus the total number of quantum parameters typically grows linearly with $N_q L$. Because the training procedure requires evaluating the quantum circuit once per collocation point, the total quantum evaluation cost scales approximately as $N_{\text{col}} N_q L$, where N_{col} denotes the number of collocation points. When using a QNN-based embedding, additional quantum evaluations are required to compute the embedding angles, contributing a cost proportional to the depth of the embedding circuit. In contrast, classical embeddings incur no quantum overhead, but require additional backpropagation through a classical network.

Algorithm 1 Training Algorithm for the QNN-TE-QPINN Framework (Navier–Stokes)

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1: Input:
   Navier–Stokes momentum operators  $\mathcal{R}_x[\cdot], \mathcal{R}_y[\cdot]$ 
   Boundary conditions: no-slip walls and lid-driven boundary
   Pressure reference constraint  $p(0,0) = 0$ 
   Collocation sets  $\mathcal{S}_{\text{int}}, \mathcal{S}_{\text{wall}}, \mathcal{S}_{\text{lid}}$ 
   Boundary weight  $\lambda_{\text{B}}$ 
   Maximum iterations  $N_{\text{max}}$ , tolerance  $\varepsilon$ 
   Embedding type: QNN-based
2: Initialization:
3: Initialize quantum embedding parameters  $\theta_Q$ 
4: Initialize variational circuit parameters  $\theta_{\text{var}}$ 
5: Select readout observables  $O_p, O_\psi$ 
6: Normalize spatial coordinates  $(x,y)$  to  $(\tilde{x}, \tilde{y}) \in [-1, 1]^2$ 
7: Initialize optimizer for  $(\theta_Q, \theta_{\text{var}})$ 
8: while not converged do
9:   Forward evaluation
10:  for each collocation point  $(x^j, y^j)$  do
11:    Compute embedding angles  $\alpha(x^j, y^j; \theta_Q)$ 
12:    Prepare encoded quantum state  $|\psi_{\text{enc}}\rangle = U_{\text{enc}}(\alpha) |0\rangle^{\otimes N_q}$ 
13:    Apply variational circuit  $|\psi_{\text{var}}\rangle = U_{\text{var}}(\theta_{\text{var}}) |\psi_{\text{enc}}\rangle$ 
14:    Evaluate predicted fields  $\tilde{p}(x^j, y^j) = \langle \psi_{\text{var}} | O_p | \psi_{\text{var}} \rangle$ ,
       $\tilde{\psi}(x^j, y^j) = \langle \psi_{\text{var}} | O_\psi | \psi_{\text{var}} \rangle$ 
15:    Compute velocity components  $u = \frac{\partial\tilde{\psi}}{\partial y}$ ,  $v = -\frac{\partial\tilde{\psi}}{\partial x}$ 
16:    Compute all spatial derivatives of  $(u, v, p)$  via hybrid
      differentiation
17:  end for
18:  Loss evaluation
19:  Compute PDE residual loss  $\mathcal{L}_{\text{PDE}}$ 
20:  Compute no-slip loss  $\mathcal{L}_{\text{wall}}$ 
21:  Compute lid-driven boundary loss  $\mathcal{L}_{\text{lid}}$ 
22:  Compute pressure reference loss  $\mathcal{L}_{\text{ref}} = p(0,0)^2$ 
23:  Total loss:  $\mathcal{L} = \mathcal{L}_{\text{PDE}} + \lambda_{\text{B}} (\mathcal{L}_{\text{wall}} + \mathcal{L}_{\text{lid}} + \mathcal{L}_{\text{ref}})$ 
24:  Gradient computation
25:  Compute  $\nabla_{\theta_Q} \mathcal{L}$  via hybrid differentiation (classical backprop
      through the embedding network and quantum parameter-shift for
      embedding gates)
26:  Compute  $\nabla_{\theta_{\text{var}}} \mathcal{L}$  via quantum parameter-shift rule
27:  Parameter update
28:  Update  $(\theta_Q, \theta_{\text{var}})$  with selected optimizer
29: end while
30: Output:
31: Trained parameters  $(\theta_Q^*, \theta_{\text{var}}^*)$  and learned fields  $\tilde{p}(x,y)$  and
       $\tilde{\psi}(x,y)$ 

```

The classical computational cost includes evaluating the PDE residuals, computing all first- and second-order spatial derivatives via automatic differentiation, enforcing the boundary and reference constraints, and computing the loss gradients. These classical operations scale with N_{col} and with the complexity of the derivative chain-rule structure in the Navier–Stokes system. Although the Navier–Stokes residuals are more expensive to evaluate than those of scalar PDEs, the cost remains polynomial in the number of collocation points.

The parameter count of the model grows polynomially with N_q , L , and the depth of the embedding circuit. This ensures that the hybrid quantum architecture remains trainable with standard gradient optimizers, while allowing the expressive capacity of the model to be tuned according to the available quantum resources. The separation between embedding and

variational parameters further provides flexibility in distributing computational effort between classical preprocessing and quantum state preparation, making the architecture compatible with near-term quantum devices.

G. Representational Capacity and Parameter Efficiency

The QNN-TE-QPINN framework leverages the expressive power of parameterized quantum circuits to approximate the nonlinear solution manifold of the Navier–Stokes equations. Each variational quantum circuit acts on an N_q -qubit Hilbert space of dimension 2^{N_q} , providing a high-dimensional latent representation in which nonlinear transformations can be implemented using relatively shallow circuits. Although the model does not attempt to explore the full Hilbert space, the exponentially large state space provides structural flexibility that may be advantageous for encoding complex flow features.

The combination of a trainable quantum embedding and a shared variational circuit allows the model to adaptively map the spatial coordinates into quantum states that highlight relevant flow structures, including boundary layers, shear regions, and vortex formation. Entanglement generated by the variational layers introduces correlations across qubits, enabling the quantum model to represent multi-scale interactions and nonlinear coupling between pressure and velocity. Because the Navier–Stokes system is characterized by strongly nonlinear advection and spatial stiffness near solid boundaries, the ability of quantum circuits to implement global transformations in a compact form may offer a representational advantage over comparable classical networks.

At the same time, the architecture remains compatible with near-term quantum devices through the use of hardware-efficient ansätze and shallow embedding circuits. The model balances expressiveness and resource cost by restricting parameter growth to linear scaling in N_q and L , rather than relying on deep or highly entangled circuits. As quantum hardware improves, this structural expressivity may support more accurate and efficient quantum-assisted solvers for fluid dynamics and other nonlinear PDEs, while preserving a modular hybrid design that integrates smoothly with classical automatic differentiation and physics-informed optimization.

IV. NUMERICAL RESULTS

In this section, we evaluate the proposed QNN-TE-QPINN framework for the steady incompressible Navier–Stokes equations associated with the lid-driven cavity problem. The experimental study is designed to assess the training behavior, solution quality, and generalization capability of the proposed quantum-assisted model. We first describe the experimental setup, including the computing environment, model configurations, quantum circuit architectures, collocation strategy, and reference solutions. We then compare different embedding strategies, analyze the training behavior against classical and hybrid quantum baselines, and report inference results using quantitative error metrics.

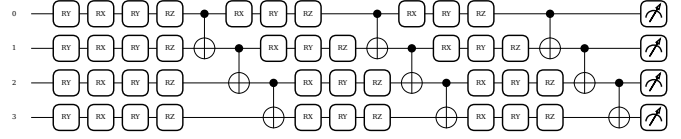


Fig. 2. Illustrative hardware-efficient VQC architecture with four qubits and three layers. In the main experiments, the number of variational layers is specified separately for each configuration.

Experimental Setup: All experiments were conducted on the REmotely-managed Power Aware Computing Systems and Services (REPACSS) high-performance computing platform [10]. Classical neural network components were implemented using PyTorch [11], while quantum circuits were constructed and simulated using PennyLane [12]. Due to current limitations in quantum hardware availability, queue times, and the overhead associated with derivative evaluation for physics-informed training, all training experiments were performed in a simulator environment. Forward inference of the trained quantum circuits may, in principle, be executed on quantum hardware, although full physics-informed training on hardware would require additional considerations related to measurement noise, shot complexity, and derivative estimation.

A classical PINN baseline with four hidden layers and 32 neurons per layer was used for comparison, resulting in 6,594 trainable parameters for approximating the pressure and stream-function fields. In contrast, for the four-qubit configuration with ten variational layers, the FNN-TE-QPINN and QNN-TE-QPINN architectures required 608 and 360 trainable parameters, respectively. This comparison highlights the parameter efficiency of the quantum-assisted models.

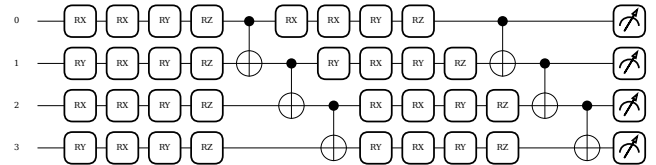


Fig. 3. Illustrative QNN-based quantum embedding circuit with four qubits and two layers.

Observable and Solver Configuration: In the experiments reported in this work, the pressure field $p(x, y)$ and the stream-function field $\psi(x, y)$ are represented by two separate variational quantum circuits with identical architectures but independent trainable parameters. Each circuit employs a Pauli- Z -based readout observable to produce one scalar output field. Consequently, two circuit evaluations are required per collocation point: one for the pressure solver and one for the stream-function solver. This design reduces unwanted coupling between the representations of pressure and stream function, simplifies the optimization procedure, and provides separate control over the approximation of each physical quantity while

preserving a common architectural template across the two solvers.

Quantum Circuit Architectures: The quantum solver uses a hardware-efficient VQC composed of single-qubit rotation gates (R_x, R_y, R_z) and CNOT entangling operations, as illustrated in Fig. 2. Each VQC acts as a physics-informed quantum solver for one output field and is applied uniformly across all collocation points.

The classical spatial coordinates are encoded into quantum states using the QNN-based embedding circuit shown in Fig. 3. The embedding circuit maps normalized coordinates to data-encoding rotation angles, which are then applied before the variational evolution. The embedding circuit uses the same number of qubits as the VQC, and its trainable parameters are optimized jointly with the variational circuit parameters.

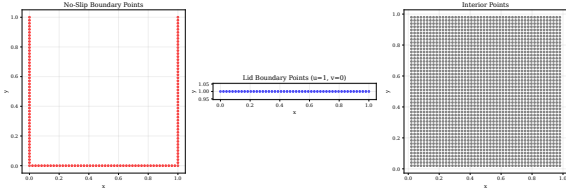


Fig. 4. Distribution of collocation points for the lid-driven cavity problem, including interior points, stationary no-slip boundaries, and the moving-lid boundary.

Collocation Points and Reference Solution: The training data consist of a uniform 50×50 grid of collocation points covering the computational domain, as illustrated in Fig. 4. The collocation set includes interior points, no-slip wall boundaries on the left, right, and bottom surfaces, and a moving-lid boundary on the top surface of the cavity. This sampling strategy ensures that the governing equations and boundary conditions are enforced throughout the domain in a physics-informed manner.

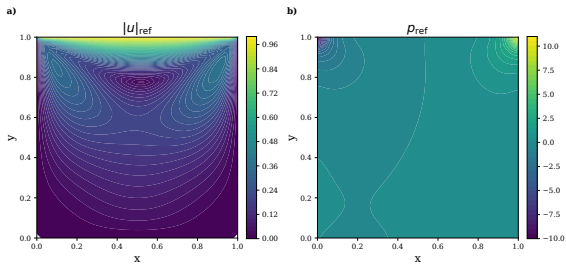


Fig. 5. Reference solutions obtained using the classical RK45 solver for (a) velocity magnitude and (b) pressure.

To evaluate the accuracy of the learned solutions, reference velocity and pressure fields were generated using a classical pseudo-time integration approach. The steady incompressible Navier–Stokes equations were written in pseudo-time form, spatially discretized on the computational grid, and integrated using an adaptive Runge–Kutta (RK45) solver until convergence to a steady state. The resulting reference solutions for the velocity magnitude and pressure distribution are shown

in Fig. 5 and are used as ground truth for quantitative error analysis during training and inference.

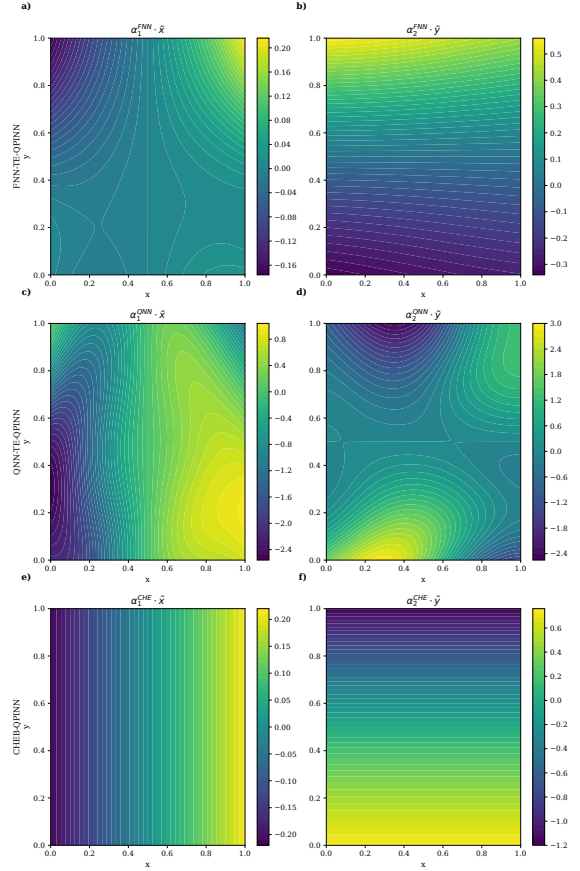


Fig. 6. Coordinate-dependent quantum encoding patterns obtained from classical collocation points for three embedding strategies: FNN-TE-QPINN, QNN-TE-QPINN, and Chebyshev-QPINN.

Embedding Strategy Comparison: Since the input data for the solver consist of classical spatial coordinates, these coordinates must be embedded into quantum states prior to variational processing. The choice of embedding strategy plays a critical role in determining the expressiveness and trainability of quantum physics-informed models. To assess the impact of embedding design, we compare three approaches: a classical feedforward neural network (FNN)-based trainable embedding, a quantum neural network (QNN)-based trainable embedding, and a fixed Chebyshev encoding.

Fig. 6 illustrates the coordinate-dependent encoding patterns produced by each method. Both the FNN-TE-QPINN and QNN-TE-QPINN embeddings yield smooth and structured representations over the computational domain, indicating that the trainable embeddings adapt the input feature map during optimization. In contrast, the Chebyshev-based embedding produces more rigid patterns dominated by predefined horizontal and vertical variations, reflecting its limited adaptability.

These results highlight the importance of adaptive, learnable embedding mechanisms in quantum-assisted physics-informed learning. While both classical and quantum trainable

TABLE I
FINAL TOTAL TRAINING LOSS FOR DIFFERENT EMBEDDING CONFIGURATIONS. QPINN MODELS USE TEN VARIATIONAL LAYERS AND FIVE EMBEDDING LAYERS, AND ALL MODELS ARE TRAINED FOR 100 EPOCHS USING THE SAME COLLOCATION SET.

Qubits	Chebyshev-QPINN	FNN-TE-QPINN	QNN-TE-QPINN
2	4.69	2.67	3.49
4	2.40	1.99	1.71
6	1.85	1.40	1.42

*Classical PINN baseline: final loss = 2.21 under the same training setting.

embeddings are capable of producing structured coordinate encodings, fixed analytical mappings such as Chebyshev-based encodings provide less flexibility for adapting the feature representation to nonlinear fluid dynamics problems.

Training Performance: We now examine the training performance of the proposed QNN-TE-QPINN solver and compare it with classical PINN and FNN-TE-QPINN baselines. All models are trained for 100 epochs using the same set of collocation points. The Reynolds number is fixed at $Re = 10$, corresponding to a kinematic viscosity $\nu = 0.1$, which leads to a relatively smooth flow regime and enables stable comparison with the RK45 reference solution.

Fig. 7 shows the evolution of the training loss and the L_2 relative error for the QNN-TE-QPINN model. Both metrics decrease smoothly with minimal oscillations, indicating stable optimization. The predicted velocity magnitude and pressure fields reproduce the main features of the reference solution, while the absolute errors remain relatively small over most of the domain, suggesting effective enforcement of the governing equations and boundary conditions.

Fig. 8 presents a direct comparison among PINN, FNN-TE-QPINN, and QNN-TE-QPINN. The classical PINN exhibits early saturation in the training loss, whereas both trainable-embedding quantum models achieve lower loss values and improved convergence behavior. Among the quantum-assisted models, QNN-TE-QPINN achieves competitive performance relative to FNN-TE-QPINN while using fewer trainable parameters.

Table I summarizes the effect of the number of qubits on the final training loss of the QPINN-based models. Since the classical PINN does not depend on the number of qubits, its values are reported only as baseline comparisons under the corresponding training and collocation settings. The Chebyshev-QPINN baseline uses a fixed analytical encoding and therefore does not learn a trainable embedding map, unlike the FNN-TE-QPINN and QNN-TE-QPINN models. The results show that the two-qubit QNN-TE-QPINN configuration has limited representational capacity, leading to a higher loss than the PINN and FNN-TE-QPINN baselines, although it still outperforms the Chebyshev-QPINN configuration. Increasing the number of qubits to four substantially improves the QNN-TE-QPINN performance, where it achieves the lowest loss among all reported configurations. For six qubits, both trainable-embedding QPINN models achieve similar losses, with FNN-TE-QPINN slightly outperforming QNN-TE-QPINN. These results suggest that the QNN-based trainable embedding is

competitive and can provide strong parameter-efficient performance, particularly in the four-qubit configuration.

Table II presents the final training loss under different Reynolds numbers. The Reynolds number measures the ratio of inertial to viscous effects, $Re = \frac{\rho UL}{\mu} = \frac{UL}{\nu}$. At low Reynolds numbers, viscous diffusion dominates and the flow field is relatively smooth. As Re increases, the influence of the nonlinear convective terms becomes stronger, boundary layers become sharper, and the approximation problem becomes more challenging for both PINN and QPINN solvers. The results show that QNN-TE-QPINN performs competitively across the tested Reynolds numbers. For $Re = 100$ and $Re = 1000$, its performance is comparable to FNN-TE-QPINN, while for $Re \geq 3000$ it achieves the lowest final training loss among the three models. These findings suggest that the QNN-based trainable embedding may be beneficial in more convection-dominated regimes, although further experiments are needed to establish robust scaling trends.

TABLE II
COMPARISON OF FINAL TOTAL TRAINING LOSS FOR PINN, FNN-TE-QPINN, AND QNN-TE-QPINN UNDER VARYING REYNOLDS NUMBERS. THE QPINN MODELS USE SIX QUBITS, AND ALL MODELS ARE TRAINED FOR 100 EPOCHS USING THE SAME COLLOCATION SET.

Reynolds Number	PINN	FNN-TE-QPINN	QNN-TE-QPINN
100	1.22	1.19	1.21
1000	1.32	1.12	1.12
3000	1.21	1.19	1.10
5000	1.21	1.19	1.11
10000	1.19	1.19	1.10

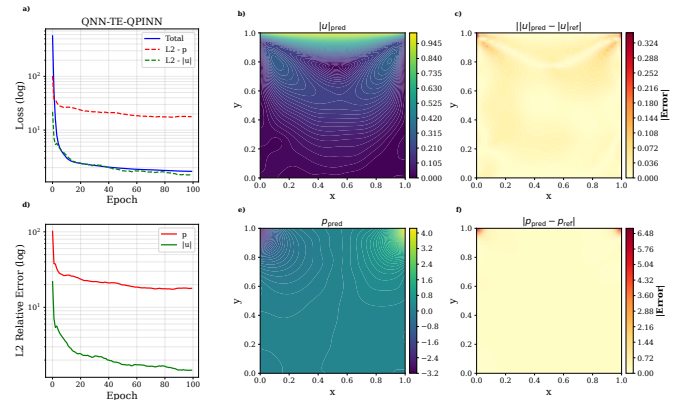


Fig. 7. Training behavior of the QNN-TE-QPINN model after 100 epochs: (a) total loss, (d) L_2 relative error; (b) and (e) model-predicted velocity magnitude and pressure fields; (c) and (f) absolute error with respect to the RK45 reference solution.

The training results indicate that the QNN-TE-QPINN model can be trained with relatively small quantum circuits, with the four- and six-qubit configurations providing the most effective performance in the present experiments. Increasing the number of qubits can improve expressive capacity and may allow comparable performance with shallower variational circuits, which is advantageous for NISQ devices where circuit depth is limited by noise and decoherence. In practice, how-

ever, the experiments are constrained by the overhead of classical quantum simulation, limiting the reported configurations to at most six qubits or fewer than twenty variational layers, even on high-performance computing platforms equipped with modern GPUs such as the NVIDIA H100. The hardware-efficient circuit architecture employed in these experiments is illustrated in Fig. 2.

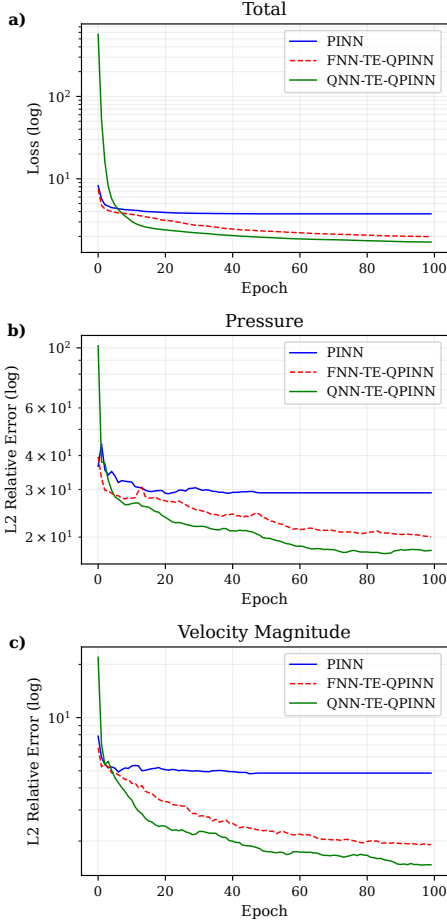


Fig. 8. Training performance comparison of PINN, FNN-TE-QPINN, and QNN-TE-QPINN: (a) total loss, (b) L_2 relative error of the pressure field p , and (c) L_2 relative error of the velocity magnitude u .

Inference Results: Following training, the generalization capability of the QNN-TE-QPINN model was evaluated on previously unseen evaluation points. Fig. 9 illustrates the inferred velocity magnitude and pressure fields, together with their corresponding error distributions relative to the RK45 reference solution.

The inferred velocity magnitude shows good agreement with the reference solution, achieving a mean squared error (MSE) of 6.64×10^{-4} , an L_2 relative error of 9.71×10^{-2} , and a maximum absolute error of 3.05×10^{-1} . The pressure field exhibits larger errors, with an MSE of 1.61×10^{-1} , an L_2 relative error of 5.51×10^{-1} , and a maximum absolute error of 6.02. The larger pressure error is consistent with the difficulty of accurately recovering pressure in incompressible Navier–

Stokes solvers, where pressure is coupled to the velocity field through the momentum equations. In physics-informed learning frameworks, errors in spatial derivatives of the learned stream function can propagate into the pressure residuals, making pressure reconstruction more sensitive than velocity prediction. Nevertheless, the predicted pressure field captures the main global structure of the reference solution. Improving pressure accuracy remains an important direction for future work and may require specialized loss weighting, auxiliary constraints, or mixed formulations that treat pressure and velocity asymmetrically.

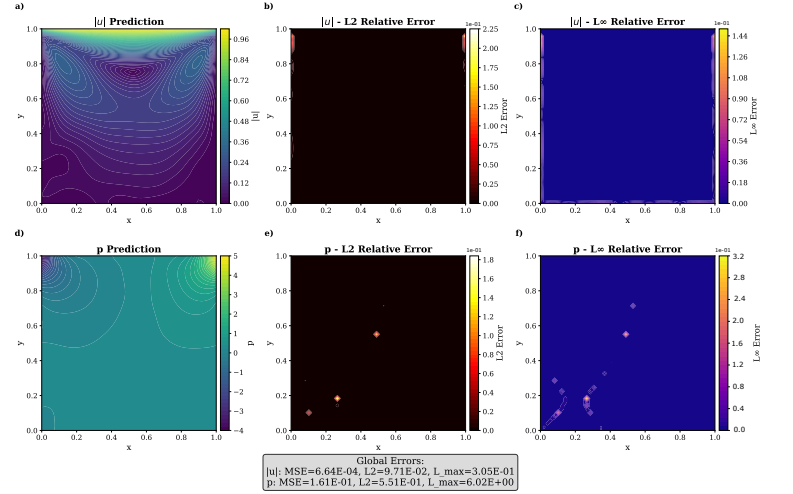


Fig. 9. Inference results of the QNN-TE-QPINN solver after training for 100 epochs. (a) and (d) Predicted velocity magnitude $|u|$ and pressure p ; (b) and (e) error distributions for $|u|$ and p ; (c) and (f) maximum-error visualizations with respect to the RK45 reference solution.

V. CONCLUSION

In this work, we proposed a QNN-based trainable-embedding QPINN framework for the steady incompressible Navier–Stokes equations in the lid-driven cavity problem. By learning data-encoding parameters within a quantum model, the proposed QNN-TE-QPINN provides adaptive, problem-dependent quantum embeddings compatible with near-term quantum architectures. Numerical results showed stable training and competitive accuracy compared with classical PINNs and hybrid QPINNs using classical embeddings, while requiring fewer trainable parameters. The best performance was obtained with four qubits and ten variational layers, where QNN-TE-QPINN improved over the classical PINN and performed comparably to FNN-TE-QPINN. However, the fully quantum embedding did not show a consistent advantage across all model sizes, emphasizing the need for resource-aware architectural design. The results indicate that QNN-based embeddings are a viable alternative for quantum-assisted physics-informed learning. Future work will study their scaling behavior, improve optimization strategies, and address gradient pathologies such as barren plateaus.

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